

EXECUTIVE SUMMARY

USC BDHSC Student Case Competition – Team HAC

Research Question

This study examines how surgery and conservative treatments affect patients differently using electronic health record data. The goal is to support clinical decision making when trial evidence is limited or does not reflect the patients seen in real practice, by estimating patient-specific treatment effects rather than population averages.

Motivation

Treatment decisions in the dataset were made by clinicians rather than through random assignment. Only 24 percent of patients received surgery, creating a large imbalance between treatment groups. Because patients differed before treatment, direct comparisons of outcomes would reflect these differences rather than the effect of surgery itself. A causal inference framework was therefore required to adjust for selection bias and estimate treatment effects as if patients had been randomly assigned.

Exploratory Analysis

Early data exploration revealed three major challenges: uneven treatment assignment, skewed and bimodal outcome distributions, and substantial missing clinical data. These findings ruled out standard linear methods and guided the use of propensity-based adjustment, outcome transformations, and principled imputation instead of deleting incomplete cases.

Feature Selection and Engineering

Clinical variables were standardized to allow fair comparisons across patients, and body mass index was calculated from height and weight when missing. Feature selection focused on clinically relevant baseline information while avoiding variables that could introduce post-treatment bias. An uplift-aligned F-filter was applied to prioritize features that contribute to differences between treatment and control outcomes, ensuring the model focused on learning treatment effect heterogeneity rather than outcome prediction alone. Missing values were imputed using healthcare-supported methods, with indicators included to capture systematic missingness. Cost and time outcomes were log-transformed to improve numerical stability and model performance.

Validity of Estimates

To address selection bias, inverse probability weighting based on propensity scores was applied, resulting in substantially improved balance between treatment groups. Covariate balance diagnostics confirmed comparability across baseline clinical features. Treatment effects were then estimated using multiple causal meta-learners, including S, T, X, and R learners, as well as a neural-network-based T learner. Comparing multiple estimators reduced reliance on any single modeling assumption and strengthened robustness.

Model Evaluation and Results

Models were evaluated using five-fold cross-validation and cross-fitted R-loss, a metric specifically designed to assess the accuracy of individualized treatment effect estimation. Using identical data splits ensured fair model comparison. The S learner consistently performed best across all outcomes, with R-loss values around 0.0036–0.0038 for Y1 through Y3, 0.0216 for Y4, and 2.24 for Y5. Predictive accuracy was balanced across treatment groups, with errors near 0.05 for clinical outcomes and between 25 and 29 for cost outcomes, supporting the reliability of estimated treatment differences.

Clinical Heterogeneity

Estimated treatment effects varied widely across patients, demonstrating that surgery does not provide uniform benefit. Baseline clinical characteristics were associated with meaningful differences in pain relief, recovery time, functional improvement, and cost. These findings highlight the limitations of average treatment effects and underscore the need for patient-specific analysis in clinical decision making.

Implications

Final outputs include patient-level treatment effect estimates, clinical driver analyses, and a CDS-ready table suitable for integration into care workflows. By adjusting for selection bias, rigorously validating models, and explicitly modeling heterogeneity, this approach supports more informed, personalized treatment decisions and contributes to the development of a learning healthcare system grounded in real-world evidence.